

FORMULATION AND EVALUATION OF MULTIPURPOSE HERBAL CREAM

A PROJECT REPORT

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CERTIFICATE

This is to certify that project work embodied in this report entitled “ **Formulation and Evaluation of Multipurpose Herbal Cream**” was carried out by Ms. Aauji Pooja Padambahadur , Ms. Apani Kavita Hasmukhbhai , Ms. Bhavsar Palak Harishkumar , Ms. Chaudhari Arpitakumari Dilipbhai , Mr. Dobariya Parthkumar Gobarbhai under my guidance in partial fulfilment for the degree of Bachelor of Pharmacy in 8th semester of Gujarat Technological University, Ahmedabad during the academic year 2022-23.

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DECLARATION

We hereby declare that the Project report submitted along with the Project entitled **“Formulation and Evaluation of Multipurpose Herbal Cream”** submitted in partial fulfilment for the degree of Bachelor of pharmacy in Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by us at Government Pharmacy College, Surat under the supervision of Dr. Kalpesh A. Patel and that no part of this report has been directly copied from any students’ reports or taken from any other source, without providing due reference.

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ABSTRACT:

Herbal medicinal products are most preferable and safer with very less side effects than the synthetic products. Now a days there is a increases in demand of herbal formulations in the world pharmaceutical market. The main objective of this work is to formulate and evaluate a cosmetic preparation multipurpose face cream was made from herbal ingredients. The cream was prepared by using the cream base that is bee's wax, liquid paraffin, borax, distilled water, rose oil, Aloe Vera gel, Neem extract and Orange Peel Powder. The cream was prepared by geometric and homogenous mixing of all the excipients and the herbal extracts. By using this technique, we have developed three batches of our herbal cream, namely F1, F2, and F3. All three batches were evaluated for different parameters like color, odour, texture, Irritancy, Washability , pH, Phase separation, Spread ability. The results indicates that formulated cream is stable, effective, and therefore, can be tested further for its performance. Thus, in the present work, we found good quality of herbal face cream, with their beneficial effects.

Key words:

Herbal Cream, Turmeric, Neem, Aloe-Vera, Cosmetic, Evaluation, Orange Peel, Skin, Aqueous And Alcoholic Extracts.

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INTRODUCTION:

The skin is the largest organ of the body, with a total area of about 20 square feet. The skin protects us from microbes and the elements, helps regulate body temperature, and permits the sensations of touch, heat, and cold.

Skin has three layers:

- The epidermis**, the outermost layer of skin, provides a waterproof barrier and creates our skin tone.
- The dermis**, beneath the epidermis, contains tough connective tissue, hair follicles, and sweat glands.
- The deeper subcutaneous tissue (**hypodermis**) is made of fat and connective tissue.

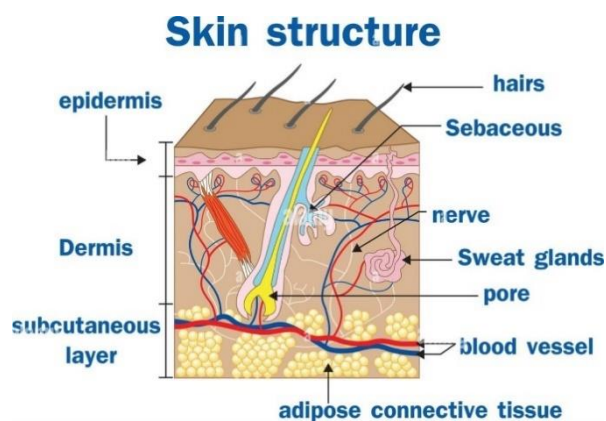


Fig. 1 – Skin Structure

Functions of Skin:

- 1. Protection:** skin act as an anatomical barrier against pathogens and prevent damage between the internal and external environment. Langerhans cells in the skin are part of the adaptive immune system.
- 2. Sensation:** It involves senses of heat and cold, touch, pressure, vibration, and tissue injury.
- 3. Heat Regulation:** The skin contains a blood supply far greater than its requirements which allows precise control of energy loss by radiation, convection, and conduction.

Dilated blood vessels increase perfusion and heat loss, while constricted vessels greatly reduce cutaneous blood flow and conserve heat.

4. Control of Evaporation: The skin provides a relatively dry and semi – impermeable barrier to fluid loss. Loss of this function contributes to the massive fluid loss in the burns.

5. Storage and Synthesis: Act as a storage centre for water and lipids, as well as the means of synthesis of vitamin D by action of UV on many parts of the skin.

6. Water Resistance: The skin act as a water resistance barrier so essential nutrients are not washed out of the body.

There are quality cosmetics products available in the market which includes: skin care creams, lotions, perfumes, lipsticks, nail polish, eye and facial makeup, hair sprays, bath oils, bubble baths, bath salts, powders, hand sanitizer, body wash and many others. Among these products, Cream is the topical drug that is suggested the most.⁴

Compared to synthetic creams, herbal creams provide several benefits. Now a days the demand of herbal cosmetics are increasing day by day. Herbal formulations are receiving more concentration in public because of their high-quality properties and less side effects. Additionally, this also provides skin with necessary nutrients.¹ Herbal extracts are used in the cosmetic preparations for augmenting beauty and attractiveness. Herbal cosmetics are classified on the basis of dosage form like - cream, powder, soaps, solutions, etc. and according to part or organ of the body to be applied for like; cosmetics for skin, hair, nail, teeth and mouth etc.²

1.1 Definition of Cream:

Cream is defined as semisolid emulsions which are oil in water (o/w) or water in oil (w/o) type and these semisolid emulsions are intended for external application.¹

1.1.1 Types of cream:

- (i) Oil In Water (O/W)
 - (ii) Water In Oil (W/O)
-

- (i) **Oil In Water (O/W)** creams are a type of emulsion in which the oil is dispersed as droplets throughout the aqueous phase.
- (ii) **Water In Oil (W/O)** creams are a type of emulsion in which the water is dispersed as droplets throughout the oily phase.

1.1.2 CLASSIFICATION OF CREAMS:

All the skin creams can be classified on different basis:

1. According to function, e.g. cleansing, foundation, massage, etc.
2. According to characteristics properties, e.g. cold creams, vanishing creams, etc.
3. According to the nature or type of emulsion, e.g Oil-in-Water (O/W) and Water-in-Oil (W/O)

1.2 Introduction of cream:

It is applied on outer part or superficial part of the skin and its main ability is to remain for a longer period of time at the site of application. The function of a skin cream is to protect the skin against different environmental condition, weather and gives soothing effect to the skin.³

Creams are used for many purposes such as protective, cleansing, improving appearance, beautifying, or for therapeutic activity. Mainly topical formulations are used for the delivery of drug into the underlying skin layer or mucous membrane of the skin, for the better delivery of drug at the specific site of action to treat various skin disorders. Unmedicated and medicated creams are mostly used for the treatment of various skin conditions and dermatoses. cream contains one or more drug substances dispersed or dissolve in a suitable base.⁴

A large quantity of creams is existing in the market under the make of natural, protected, organic, and herbal. Most of the creams at present existing in the market use the synthetic polymers, emulsifiers, perfuming agents, pigments, surfactants and thickeners to form the base. There is wide need to substitute toxic synthetic agent from base using natural agents.¹

1.3 Ingredients:

The natural ingredients chosen for preparation of herbal cream are aloe-vera gel, orange peel powder and neem powder. The choice of these ingredients is based on their individual properties. Aloe Vera gel is used as a moisturizer, to reduce pimples and acne and also used for treatment of burn wounds. Neem is used as an antifungal and anti-inflammatory and it is also used to reduce scar, pigmentation, redness and itching of the skin. orange peel powder helps in removing oil from the skin and makes skin look smooth, firm, and soft: due to the high content of vitamin C it gives smoothen textures to skin.

1.3.1 Pharmacognosy of Turmeric:^{[11] [12]}

Scientific name : *Curcuma longa*

Synonyms :- Saffron Indian; haldi (Hindi); Curcuma; Rhizoma cur-cumae.

Family: Zingiberaceae.

Biological Source :- Turmeric is the dried rhizome of *Curcuma longa* Linn. (syn. *C. domestica* Valetton).

Chemical constituents:

1. Curcuminoids- Non-volatile colouring matter.
2. Curcumin, a diferuloylmethane; desmethoxy dicinnarmoylmethane; bidesmethoxy curcumin.



Fig : 2 Turmeric

Benefits of Turmeric:

- 1) It has anti-inflammatory, antimicrobial and antioxidant properties that help to manage several skin issues like signs of aging, pigmentation, acne, etc.
- 2) Turmeric has a bioactive component called curcumin which makes it anti-inflammatory. The key health benefits of turmeric come from this curcumin.
- 3) Curcumin being a high antioxidant compound boosts collagen production and strengthens the elasticity of skin.

- 4) Turmeric is considered to be a potent ingredient for skin lightening and revealing glow. It is a powerful antioxidant ingredient that can reduce dark spots, blemishes and hyperpigmentation.
- 5) Being anti-microbial and anti-inflammatory, turmeric can kill the bacteria that causes psoriasis and reduce skin inflammation.

1.3.2 Pharmacognosy of Neem:^[13] ^[14]

Synonyms: Hin.-Nira, nimb; Mal. – Veppa; Mar. – Limba, Oriya- Nimba; Tam- Vembu.

Scientific name : Azadirachta indica

Biological Source: Neem consists of the fresh or dried leaves and seed oil of Azadirachta indica J. Juss (Melia Indica or M. azadirachta Linn.).

Family: Meliaceae.

Chemical Constituents: Azadirachtin and the others are **nimbolinin, nimbin, nimbidin, nimbidol, sodium nimbinate, gedunin, salannin, and quercetin.**



Fig: 3 - Neem

Benefits of Neem :

- 1) Neem is able to balance oil production, heal wounds, stimulate collagen formation, reduce post-acne scars and minimise skin inflammation. It works as a natural alternative to heal the skin tissue from within and simultaneously reduces hyperpigmentation and scars.
- 2) In addition to reducing marks left behind from the ghosts of breakouts past, neem also works on actively treating pimples with its antiseptic and antibacterial properties, thus earning a regular spot in anti-acne cleansers, creams and treatments.
- 3) Those suffering from sun damage and the residual skin redness will find relief in this herb's anti-inflammatory properties too. The presence of essential fatty acids in the oil derived from its leaves makes for a handy moisturiser that sinks deep within the skin. Regular usage of neem oil increases the skin's ability to absorb other subsequent active ingredients, thus helping you maximise the impact of all components in your skincare regimen.
- 4) As the skin ages, the production of collagen starts decreasing, which in turn leads to the appearance of wrinkles. Neem comes loaded with Vitamin C, which helps boost the natural production of collagen and flood the skin with antioxidants to smooth fine lines.

1.3.3 Pharmacognosy of Orange Peel:^{[15] [16] [17]}

Scientific name : Citrus x aurantium

Synonyms : Citrus vulgaris, Citrus bigaradia, Citrus aurantium amara, Biga-rade orange, Bitter orange, Seville orange, (Sweet) Portugal orange, China orange, Citrus dulcis, Cortex aurantii amar L, Seville orange peel.

Biological Sources : The orange peel is the fresh or dried outer part of the pericarp of Citrus aurantium Linn,

family : Rutaceae.



Fig. 4 – Orange peel



Fig. 5 – Orange peel powder

Benefits of orange peel :

1) Clears the skin

Orange peel exhibits excellent anti-inflammatory properties. They can suppress inflammation like a natural source of indomethacin, an anti-inflammatory drug. Also, the flavonoids in orange peels are known to permeate membranes and heal inflammation.

2) Exfoliates dead cells

The rind of orange is considered a boon for the skin as it treats blackheads and blemishes. It works as an effective exfoliating agent that helps in removing blackheads and unclogging the skin pores. Dried orange peel beads act as natural exfoliators and remove dead cells and blackheads gently and naturally, bringing a glow to the skin.

3) Brightens up

The peels will bring a bright glow to face. In addition, they help to lighten the dark spots. If anyone suffering from acne scars, dark spots or even pigmentation, orange peel powder's rich content of citric acid will take care of the issue.

4) Reverse ageing

Orange peel powder contains enough calcium that renews and regenerates the skin. It is also known to regulate sebum production, thus getting rid of the skin's excess

oil. It protects skin from sun damage with its anti-inflammatory properties. Not only this, they make skin more firm and elastic by improving collagen production. Thus, reversing the effects of aging.

5) Heals acne

Due to its cleansing, anti inflammatory, antibacterial and antifungal properties, orange peels work well against pus filled pimples and acne. They dry up the pimples allowing them to heal faster.

1.3.4 Pharmacognosy of Aloe Vera:^{[18] [19] [20] [21]}

Scientific name: Aloe barbadensis miller

Family: Asphodelaceae

Synonyms: Aloe, Musabbar, Kumari.

Biological source: Aloes are the dried juice obtained by transversely cut leaves of various species of Aloe barbedensis Miller (known as Curacao aloes), Aloe perryi Baker (Socotrine aloes), Aloe spicata Baker, and Aloe Africana Miller (Cape aloes).

Chemical constituents : Aloins, Barbaloin, β -barboloin and Isobarbaloin, which constitute the so-called 'crystalline' Aloin, present in the drug at from 10 to 30%.



Fig. 6 – Aloe Vera

Benefits of Aloe Vera:

- 1.) **Helps soothe sunburn:** Aloe Vera gel has cooling properties and is anti-inflammatory. Hence, it is one of the most natural remedies for sunburn or burnt skin. Applying this gel helps with a protective layer for the skin. It is rich in antioxidants and minerals that boost the healing process.
- 2.) **Helps to moisturize the skin:** it unclogs the pores and softens the skin. It also helps treat dry skin.
- 3.) **Fights skin-ageing:** Aloe Vera gel has vitamin C and E, beta-carotene in abundance. Therefore, it has anti-ageing properties. It also contains antimicrobial properties and is anti-inflammatory.
- 4.) **Lightens blemishes on the face:** Aloe Vera contains powers to boost skin cell reproduction, reduce redness, and fight skin inflammation, it is a natural treatment for stretch marks and acne marks. To treat freckles and lighten age spots, add some lemon juice to the gel mixture.

1.3.5 Pharmacognosy of Bees Wax : ^[22] ^[23] ^[24]

Scientific name: cera alba

Synonyms: White beeswax, yellow beeswax, cera alba, and cera flava.

Biological Source: Beeswax is the purified wax obtained from honeycomb of hive bee, *Apis mellifera* Linn and other species of *Apis*,

Family: Apidae.

Chemical constituents:

- (i) Lipids like wax- myricyl palmitate (80%).
 - (ii) Wax -acid such as cerotic acid (15%).
 - (iii) Aromatic substances: Cerolein.
 - (iv) Cholesteryl ester.
 - (v) Lignoceric acid.
 - (vi) Myristolactone.
-

- (vii) Myricyl alcohol and ceryl alcohol



Fig. 7 – Bees Wax

Benefits of Bees Wax:

- 1) Beeswax forms a protective barrier on the skin's surface, protecting skin from environmental irritants and harsh weather. Beeswax is particularly exceptional being non-comedogenic (non-pore blocking), allowing the skin to breathe freely. This makes beeswax particularly beneficial for acne-prone and sensitive skin.
- 2) Beeswax has fantastic skin-softening properties and enhances skin elasticity, helping reduce the signs of ageing.
- 3) Beeswax is anti-allergenic, anti-inflammatory, anti-oxidant, anti-bacterial and germicidal. These properties make beeswax a highly stable base ingredient.

1.3.6 Liquid Paraffin : ^[25] ^[26]

Liquid paraffin (petrolatum) is a highly refined distilled fraction of petroleum that contains a mixture of liquid saturated hydrocarbons Sweetman and used as bases and lubricants.

Liquid Parafin is used for the dry skin. It relieves dry skin conditions such as eczema, ichthyosis, and pruritus of the elderly. It works by preventing water loss from the outer layer of skin. This relieves dryness and leaves the skin soft and hydrated.

1.3.7 Borax:^[28] ^[29]

Borax, also known as sodium tetraborate, is a mineral salt commonly used in cosmetic products for years to remove excess oil, dirt, dead skin, and bacteria.

Chemical formula : $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

Molecular weight : 381.37 g/mol

Solubility : 520g/L in water at 100° C

In addition, it inhibits or reduces bacterial growth in products such as creams, bath bombs, salts, shower gels, body lotions, and shampoos. This makes it an ideal ingredient for use as a preservative or buffering agent in these products.

1.3.8 Rose oil:^[30]

Rose oil is an essential oil that's extracted from the petals of R. Damascena variety of rose plants.

Rose oil is often referred to as the “queen of all oils” because of its lovely aroma and powerful topical properties. The substance is packed with vitamins, minerals and antioxidants that help improve the health and appearance of the skin.

Benefits of Rose oil :

1) Antibacterial Properties:

Oil distilled from roses contains a wide spectrum of antibacterial properties against pathogens that cause infections. These properties help to cleanse and tone the skin while preventing the formation of acne.

2) Hydrates and Soothes Skin

Rose oil is the best face oil for dry skin. It is a natural humectant, meaning it helps to keep the skin hydrated. It does this by drawing moisture from the air and sealing it into the skin. This is perfect for those with sensitive, dry or cracked skin, as it will help to keep the skin hydrated and prevent flakiness.

3) Tones & Tightens

Rose oil is a natural astringent and helps to tone and tighten the skin. This is perfect for those with oily skin, as it will help to reduce sebum production, a leading cause of acne. Along with rose oil, our vegan skin care professionals utilize aloe, witch hazel, and arnica for our pH balancing toner to even skin tone and minimize pores

4) Combats Acne

rose oil has antiseptic and astringent properties, which makes it a more gentle way of fighting the bacteria that cause acne.

5) Quells Irritation

Rose oil has anti-inflammatory properties that nip redness in the bud, thereby soothing skin. Rose oil's ability to quell irritation also makes it a go-to product for those with inflammatory skin conditions like rosacea and eczema.

6) Boosts Healthy Collagen Production

As ageing, skin starts to produce less collagen, which leads to fine lines and wrinkles. Rose oil can help to boost healthy collagen production, which will in turn help to reduce the appearance of fine lines and wrinkles.

7) Slows Free Radical Damage

Rose oil contains antioxidants like citronellol and geraniol, helping to diminish signs of aging. The combination also slows down free radical damage, which is one of the leading causes of aging skin.

1.3.9 Vitamin E Capsule:^[31]

Vitamin E is a group of eight fat soluble compounds that include four tocopherols and four tocotrienols and found in a wide variety of foods.

Vitamin E oil's potential benefits derive from two key features: its antioxidant properties, which could fight inflammation and slow the effects of free radicals, and its moisturizing properties.

Some purported benefits of vitamin E oil include:

1) Moisturizing skin

Vitamin E is found in many moisturizers, and the oil may be used as a moisturizer to prevent or treat dry, flaking skin.

2) Reducing skin itching

Vitamin E may help to reduce itchy skin and ease eczema. Vitamin E cannot treat allergic reactions, infections, and other issues that cause skin itching. Because it

moisturizes the skin, however, it may offer temporary relief from itching caused by dry skin.

3) Preventing or treating fine lines and wrinkles

Dry skin tends to look more wrinkled than well-moisturized skin. The moisturizing benefits of vitamin E oil may help the skin look more youthful and less wrinkled.

4) Sunburn prevention

Vitamin E may help to reduce the risk of sunburn. Limited research suggests that vitamin E can prevent or reduce the formation of sunburns. Because vitamin E oil can moisturize and soothe dry, flaky skin, it may help to relieve the burning and itching that result from a sunburn.

LITERATURE REVIEW:^{[1] [6] [4] [3] [2]}

Sr. No.	Title Name	Inference	Reference
1	Formulation and Evaluation of Polyherbal Cream	Cosmetics made from either natural or synthetic components are almost in regular use universally in many different forms for enhancing the beauty. The incorporated herbs such as the dried leaves of Azadirachta indica have antibacterial, anti-inflammatory analgesic activity.	Chatur VM, Ansari NM, Joshi SK ;2022 Journal of Drug Delivery and Therapeutics
2	Formulation and Evaluation of Multipurpose Herbal Cream	Herbal cosmetics are products that are used to improve one's look. The goal of the research was to develop a herbal cream for moisturizing, nourishing, whitening, and treating various skin diseases. The selection of components is based on the agents' various therapeutic-characteristics. Various evaluation parameters are used to the cream	Somnath S Davkhar*, Aarti S Bhandari;2023 Sys Rev Pharm

CHAPTER : 2 | LITERATURE REVIEW

3	Formulation and Evaluation of Polyherbal Face Cream	The main objective of this work is to formulate and evaluate a cosmetic preparation polyherbal face cream made from herbal ingredients. extraction of dried crude herbal ingredients were done by the maceration process. After extracting the polyherbal face cream is prepared and evaluated for its various charecteristics.	Miss. Adhalrao Supriya B.1, Miss. Panmand Dipali A.2 ; 2022 International Journal of Advanced Research in Science , Communication and Technology (IJARSCT)
4	Formulation and Evaluation of Herbal Moisturizing Cream	A semi-solid product for enhancing skin tone is a moisturizing cream. Compared to synthetic creams, herbal creams provide several benefits. The aforementioned herbal cream was assessed based on factors including pH, viscosity, greasiness, washability, appearance (colour), and homogeneity by visual and tactile means. ³	Arun Kumar, Divyansh*,Neha Ansari, Rahul Shukla;2022 International Journal Of Pharmacy and Pharmaceutical Research
5	Formulation and Evaluation of Polyherbal Vanishing plus fairness expert cream	Herbal creams offer several advantages over other creams.The herbal vanishing cream was prepared by using o/w emulsion method using mixture of alcoholic extract of crude drugs including Peppermint leaves, Neem leaves, Orange peel, Ginger, Mustard, Turmeric, Almond Seed and Gram flour.The extract were used and formulated and pass all evaluation test.	Seema Y. Mendhekar , Rajnigandha R. Jori;2017 World Journal Of Pharmacy And Pharmaceutical Science s

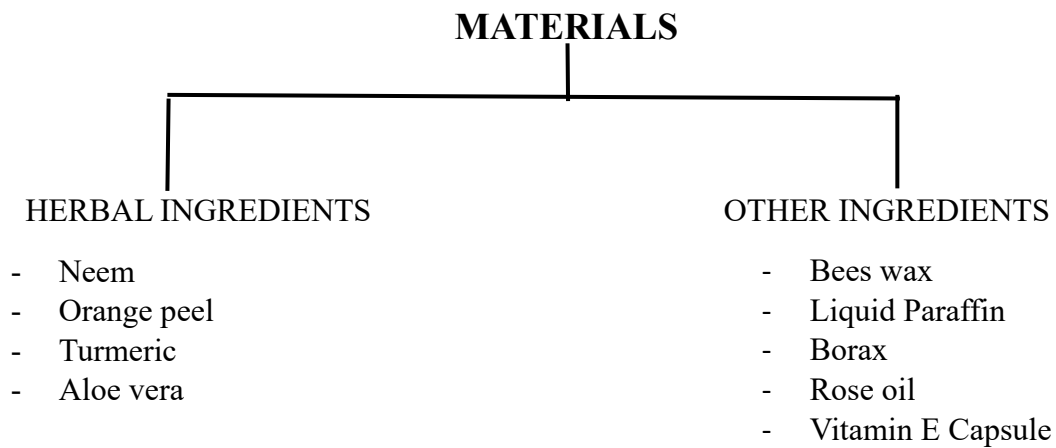
AIM & OBJECTIVE

AIM: To Formulate and Evaluate the Multipurpose Herbal Cream.

OBJECTIVE: - To Extract Phytoconstituent from Herbal Ingredients.

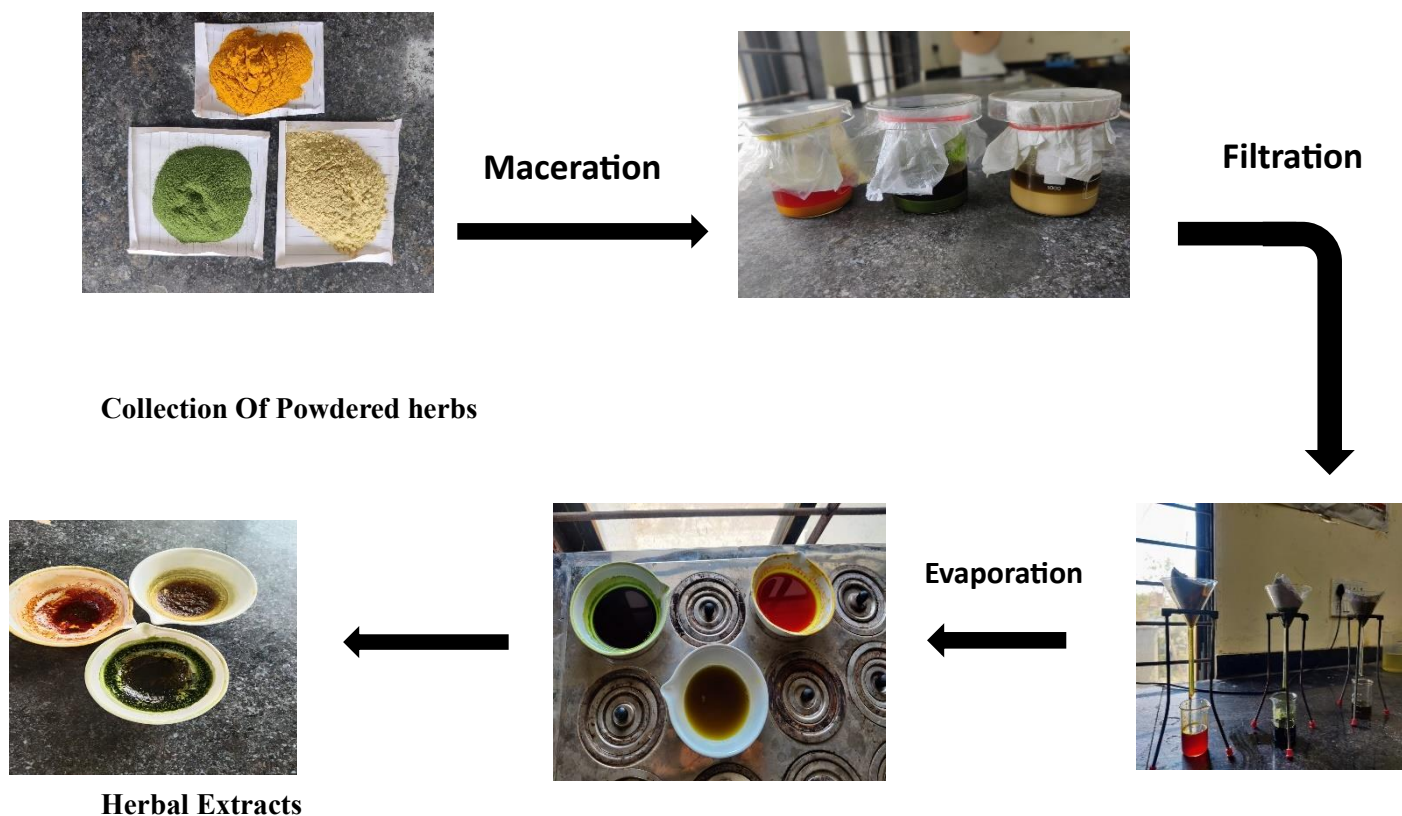
- To formulate herbal cream.
- To evaluate herbal cream.

MATERIALS AND METHODS:



METHOD OF PREPARATION:

A) EXTRACTION OF HERBAL INGREDIENTS :



B) Formulation Of Herbal Cream:

Preparation of oily phase



Preparation of Aqueous phase



Homogenous Mixing (75°C)



Herbal cream was Prepared

Fig.8 - Flow chart of Herbal cream formulation

METHODS :

For this study the maceration process is used for the extraction of herbal ingredients used in the formulation.

Maceration process : Maceration is the simple technique use for extraction of plant material. In maceration process powdered or coarse plant material is soaked in suitable solvents such as ethanol, methanol, acetone etc. It is Most commonly used inexpensive technique used for the extraction of different bioactive compounds from plant material. The selection of suitable solvent is most important factor for the extraction of particular plant extract. Maceration consist of converting large particles into smaller powdered form to increase the surface area for achieving best results from extraction by easy mixing with solvent. Then this mixture of plant material and solvent is kept for longer time (4-8 days), agitated at different intervals and filtered through a filtration medium. The efficiency for extraction depends on the type of solvent and plant material. Solvent polarity is also important factor which affects the efficiency of extraction. Due to maceration ruptures the cell structure and chemical constituents expose to react with solvent and helps in removal of active constituents from different plant materials. This method is mostly preferred for the exaction of different types of bioactive compounds at laboratory scale.³

A.) Extraction of herbal ingredients:

Approximately 15g powder of all the herbal ingredients is taken and soaked in ethanol for 6-7 days and agitated at 24 hours intervals. After maceration process the macerate is further filtered. After filtration it is subjected for evaporation to yield semi solid residue of herbal ingredients on waterbath.

B) Formulation of oily phase:

Take the liquid paraffin and bees wax in a borosilicate glass breaker at 75°C and maintain that heating temperatures (oil phase). Immediately add neem extract, turmeric extract, orange peel powder extract, Vitamin E capsule and rose oil into oily phase with continuous stirring and maintain temperature at 75° C.

C) Formulation of aqueous phase:

In other beaker, dissolve borax in distilled water by maintaining temperatures 75°C with water bath. Stir the solution with glass rod until all solid particles get dissolve. Add Aloe Vera gel in it and maintain temperature at 75°C.

D) Mixing of two phases:

The gently add heated aqueous phase in heated oily phase with continue stirring by glass rod until it forms a smooth cream.

Table 1 – Composition of Herbal ingredients based face cream:

Sr. No.	Ingredients	F1	F2	F3
1.	Turmeric extract	0.5 ml	1.0 ml	1.5 ml
2.	Neem extract	1 ml	1.5 ml	2.0 ml
3.	Orange peel powder extract	0.5 ml	1.0 ml	1.5 ml
4.	Aloe vera gel	1 g	1 g	1 g
5.	Vitamin E capsule	1 capsule	1 capsule	1 capsule
6.	Bees wax	10g	12g	15g
7.	Liq. Paraffin	14 ml	16 ml	18 ml
8.	Borax	4 g	6 g	8 mg
9.	Water	q.s	q.s	q.s
10.	Rose oil	q.s	q.s	q.s

EVALUATION PARAMETER OF CREAM: ³

- 1.) **Physical parameters:** In this test color, odour, texture, and state of cream are observed.
- 2.) **Irritancy:** Mark the area (1 cm²) on the left hand dorsal surface. Then the cream was applied to the area and after some time it is checked for its irritant effect.
- 3.) **Wash ability:** Wash ability test was carried out by applying a small amount of cream on the hand and then washing it with help of tap water.
- 4.) **Phase separation:** Prepared cream is kept in tightly closed container at room temperature away from sunlight and observed for 24 hours for phase.
- 5.) **pH:** Take 0.5 g of cream and dispersed it in 50 ml distilled water. Then check its pH by using digital pH meter.
- 6.) **Spreadability:** Spreadability is carried out for all three formulations that is, F1, F2 and F3. The less time take for the separation of the slide better the spreadability.

The spreadability was expressed in terms of time in seconds taken by two slides to slip off from the cream placed in between the slides, under certain load

$$\text{Spreadability} = \frac{m \times l}{t}$$

Where,

m = Standard weight, which is placed on the upper slide

l = length of a glass slide

t = time taken in separation of two slide

RESULT AND DISCUSSION:

The present study was undertaken to formulate and evaluate multipurpose herbal cream using ethanolic extract. The formulations were subjected to various physical evaluation parameters like physical properties, pH, phase separation, spreadability, wash ability and irritancy. The results of the same are given below in the tabular form.

1.) Physical Parameter:

Table no. 2 – Physical Parameters

Sr. No.	Parameter	Batch F1	Batch F2	Batch F3
1.	Color	Olive green	Light yellow	Light yellow
2.	Odor	Essence of rose	Essence of rose	Essence of rose
3.	Texture	smooth	Smooth	Smooth
4.	State	Semi solid	Semi solid	Semi solid

The prepared formulation cream F1, F2 and F3 on visual observations were found to be homogeneous with a pleasant odour and smooth texture.

2.) Irritancy:

Table no. 3 – Irritancy Test

Sr. No.	Parameter	Batch F1	Batch F2	Batch F3
1.	Irritancy	Non-irritant	Non-irritant	Non- irritant

The prepared formulation cream is found to be non-irritant.

3.) Wash ability :

Table no. 4 – Wash Ability Test

Sr. No.	Parameter	Batch F1	Batch F2	Batch F3
1.	Wash ability	Easily washable	Easily washable	Easily washable

The prepared formulation cream is found to be easily washable.

4.) pH :

Table no.5 – pH test

Sr. No.	Parameter	Batch F1	Batch F2	Batch F3
1.	pH	6.53	6.75	6.91

According to the results, the pH of all the three formulations that is F1, F2 and F3 were found to be in the range of 6.0 to 7.0 , which is good for skin pH. All the formulations of cream were shown pH nearer to skin required i.e. 5–7 pH.

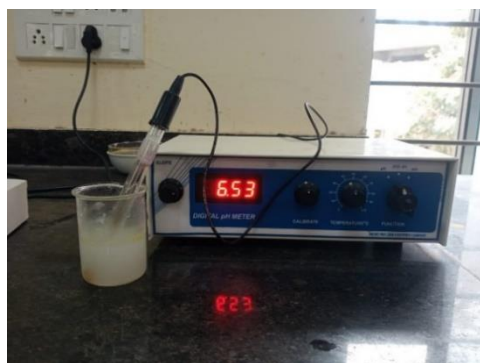


Fig. 9 - pH of F1



Fig. 10 – pH of F2



Fig. 11 – pH of F3

5.) Phase Separation:

Table no. 6 – Phase Separation Test

Sr.No	Formulation	Phase separation
1.	F1	No phase separation
2	F2	No phase separation
3.	F3	No phase separation

Prepared cream was kept in a closed container at room temperature, away from light. Then phase separation was checked for 24 h and also after 30 days. According to the results, no phase separation was observed in all three formulations.

6.) Spreadability :

Table no. 7 - Spreadability

Sr.No.	Formulation	Time (sec)	Spreadability (gcm/sec)
1.	F1	7	5
2.	F2	10	6.5
3.	F3	15	7.5

Among all the three formulations, for F1 the time taken by the two slides to separate is less so as said in the description of the evaluation test, lesser the time taken for separation of the two slides better the spreadability. So, according to this statement F1 showed desired Spreadability.

CONCLUSION:

The objective of the present work was to formulate and evaluate multipurpose herbal cream prepared by using ethanolic extract of Neem, Turmeric and orange pill powder ,vitamin E capsule and Aloe vera gel showing multipurpose effect such as whitening, anti-aging, antioxidant, antimicrobial effect.

Based on the results and discussion, the prepared formulations were found to be yellow and olive green in colour with pleasant odour and smooth texture. All the formulations had almost constant pH, homogenous, emollient, non-greasy and easily removed after the application at room temperature.

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